

# Week 09: Patterns without Labels: Clustering, Signals and Anomaly Detection

Artificial Intelligence Applications in Engineering (MUH-920 Mühendislikte Yapay Zeka Uygulamaları). Lecture notes

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*Week overview, study path and tasks:* <https://github.com/utkukose/ai-applications-in-engineering-NB-lecture/blob/main/weeks/week-09/README.md>

*Lecture page with animations and knowledge checks:* <https://utkukose.github.io/ai-applications-in-engineering-NB-lecture/weeks/week-09/lecture.html>

*Interactive lab with self-assessment and reflection:* <https://utkukose.github.io/ai-applications-in-engineering-NB-lecture/weeks/week-09/lab.html>

*Colab notebook with Python step 9 and the hands-on work:*

[https://colab.research.google.com/github/utkukose/ai-applications-in-engineering-NB-lecture/blob/main/weeks/week-09/NB09\\_unsupervised\\_signals\\_anomalies.ipynb](https://colab.research.google.com/github/utkukose/ai-applications-in-engineering-NB-lecture/blob/main/weeks/week-09/NB09_unsupervised_signals_anomalies.ipynb)

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## Overview

Sensors, logs and catalogues produce far more data than anyone can label. Unsupervised learning finds structure in such data: groups of similar records, a few directions that carry most of the variation, and points that do not fit [1, 2]. This week clusters well logs into rock types without using the geologists' labels [3], groups earthquakes by density in the catalogue of the U.S. Geological Survey [4, 5], and moves to signals: The Fourier transform and envelope analysis reveal bearing faults in vibration data, and an isolation forest learns what normal vibration looks like [6, 7, 8].

**Estimated study time:** 9 to 11 hours.

## Learning outcomes

By the end of the week, students are expected to apply k-means with proper scaling and choose the number of clusters with the silhouette score, to interpret principal components and their explained variance, to use DBSCAN for clusters of arbitrary shape and noise, to compute and read the spectrum of a sampled signal, to compute bearing fault frequencies and condition indicators such as RMS, kurtosis and crest factor, and to train and judge an anomaly detector on data from normal operation.

## Python in this week

These notes close with Python step 9: Signals, features and clustering with NumPy. It covers time vectors and sampled signals, signal statistics, windows and the axis argument, spectra with the fast Fourier transform, and k-means clustering and a z-score anomaly flag written with broadcasting, applied to machine vibration [9]. In the course materials, the step is taught at the start of the Colab notebook,

where each snippet runs in a cell next to its explanation and short exercises give immediate feedback, and later sections of the notebook mark further constructs as Python moves.

## Learning without labels

Supervised learning needs a label for every example. In engineering, labels are expensive: A geologist must interpret each metre of a well, a technician must confirm each fault, and many rare events have never been observed. Unsupervised learning works with the inputs alone. Clustering groups similar records, dimensionality reduction summarises many correlated variables with a few, and anomaly detection flags records that differ from the bulk of the data [1, 2]. The results are hypotheses that domain experts must interpret: A cluster is not a rock type, and an anomaly is not a fault, until someone checks.

## k-means and the choice of k

k-means represents each cluster by a centre and alternates two steps: assign every point to its nearest centre, then move every centre to the mean of its points. Each step lowers the sum of squared distances, so the procedure converges, although only to a local optimum [10, 11]. The k-means++ initialisation spreads the initial centres out and improves both speed and quality [12]. Because k-means uses Euclidean distance, inputs with large numerical ranges dominate unless they are scaled; gamma-ray counts in API units would otherwise swamp a photoelectric factor.

The number of clusters is a modelling choice. The within-cluster sum of squares always falls as k grows, so the elbow of its curve is only a rough guide. The silhouette of a point compares its mean distance to its own cluster with that to the nearest other cluster, and the average silhouette over all points is highest for well-separated, compact clusters [13]. Neither criterion knows the physics; well logs may form five clear clusters although geologists distinguish nine facies.

**Animation: The two steps of k-means.** Press Step to alternate between assigning points to the nearest centre and moving each centre to the mean of its points. Try several restarts: Different initial centres can end in different solutions. Run it on the [lecture page](#).

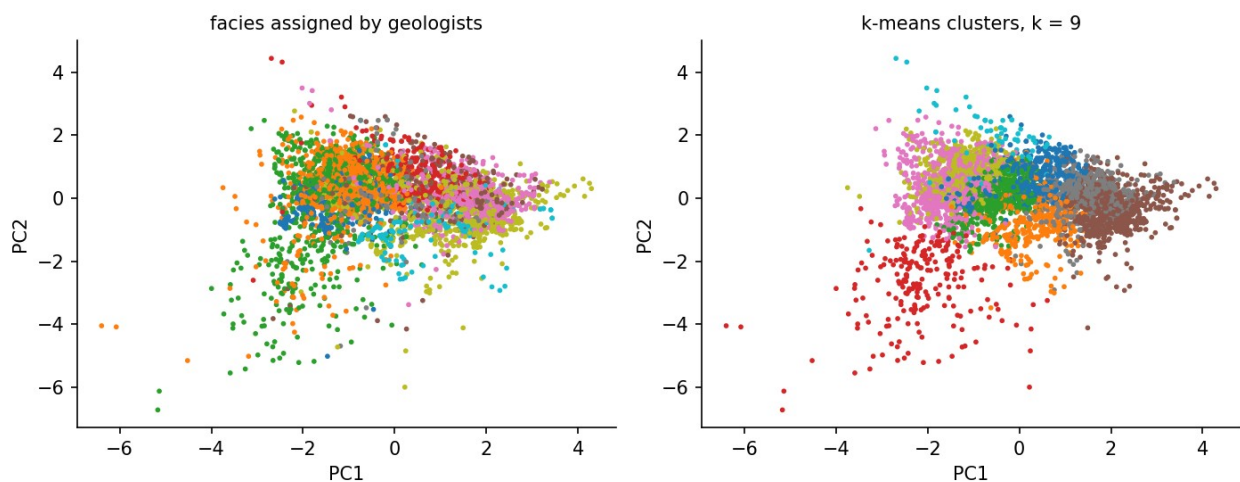


Figure 9.1. Well-log samples of the SEG facies data projected on the first two principal components, coloured by the geologists' facies (left) and by k-means clusters found without labels (right) [3].

**Check your understanding.** Why must the inputs usually be standardised before k-means?

- A. k-means cannot handle negative numbers
- B. Euclidean distances are dominated by variables with large numerical ranges
- C. Standardisation fixes the number of clusters
- D. k-means only works with values between 0 and 1

**Answer: B.** Without scaling, a variable measured in large units decides the distances and hence the clusters.

## Formulas: The k-means objective, k-means++ and the silhouette

k-means divides  $n$  points  $x_i$  into  $k$  clusters  $C_1, \dots, C_k$  with centres  $\mu_1, \dots, \mu_k$  and minimises the within-cluster sum of squares [10, 11]:

$$W = \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

The double bars denote the Euclidean length of a vector, so  $\|x_i - \mu_j\|^2$  is the squared distance. The algorithm alternates two steps, each of which can only lower  $W$ : The assignment step gives every point to its nearest centre, and the update step moves every centre to the mean of its points,

$$c(i) = \arg \min_j \|x_i - \mu_j\|^2, \mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$$

k-means++ chooses the first centre at random and every further centre with a probability proportional to the squared distance  $D(x)^2$  of a point to the nearest centre chosen so far [12]:

$$P(x) = \frac{D(x)^2}{\sum_{x'} D(x')^2}$$

The silhouette of a point compares its mean distance  $a(i)$  to the other points of its cluster with its mean distance  $b(i)$  to the points of the nearest other cluster [13]:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Values near 1 mark points deep inside their cluster, values near 0 points between clusters, and negative values points that sit closer to another cluster.

### Worked example: One iteration of k-means

Six points, A (1, 1), B (1.5, 2), C (3, 3.5), D (5, 7), E (3.5, 5) and F (4.5, 5), are clustered with  $k = 2$ , starting from  $\mu_1 = (1, 1)$  and  $\mu_2 = (5, 7)$ .

**Step 1. Assignment.** The squared distances to the two centres, for example

$$\|C - \mu_1\|^2 = (3 - 1)^2 + (3.5 - 1)^2 = 4 + 6.25 = 10.25 \text{ and}$$

$$\|C - \mu_2\|^2 = (3 - 5)^2 + (3.5 - 7)^2 = 4 + 12.25 = 16.25:$$

| Point | Coordinates | To $\mu_1$ | To $\mu_2$ | Cluster |
|-------|-------------|------------|------------|---------|
| A     | (1, 1)      | 0          | 52         | 1       |
| B     | (1.5, 2)    | 1.25       | 37.25      | 1       |
| C     | (3, 3.5)    | 10.25      | 16.25      | 1       |
| D     | (5, 7)      | 52         | 0          | 2       |
| E     | (3.5, 5)    | 22.25      | 6.25       | 2       |
| F     | (4.5, 5)    | 28.25      | 4.25       | 2       |

The sum of the smaller squared distances is  $W = 0 + 1.25 + 10.25 + 0 + 6.25 + 4.25 = 22$ .

**Step 2. Update.**

$$\mu_1 = \left( \frac{1+1.5+3}{3}, \frac{1+2+3.5}{3} \right) = \left( \frac{5.5}{3}, \frac{6.5}{3} \right) = (1.8333, 2.1667)$$

$$\mu_2 = \left( \frac{5+3.5+4.5}{3}, \frac{7+5+5}{3} \right) = \left( \frac{13}{3}, \frac{17}{3} \right) = (4.3333, 5.6667)$$

**Step 3. The new objective.** For A,

$\|A - \mu_1\|^2 = (1 - 1.83333)^2 + (1 - 2.16667)^2 = 0.694444 + 1.361111 = 2.055556$ , and in the same way the squared distances to the new centres are 2.055556, 0.138889, 3.138889 in cluster 1 and 2.222222, 1.138889, 0.472222 in cluster 2, so

$$W = 2.055556 + 0.138889 + 3.138889 + 2.222222 + 1.138889 + 0.472222 = 9.166667$$

much less than 22. A second assignment step leaves every point in its cluster, so the algorithm has converged.

### Worked example: Seeding with k-means++ and two silhouettes

**Step 1. The second centre.** With A as the first centre, the squared distances of the other points are B 1.25, C 10.25, D 52, E 22.25, F 28.25, which add up to 114. The probabilities are B 0.011, C 0.0899, D 0.4561, E 0.1952, F 0.2478, with the cumulative values 0.011, 0.1009, 0.557, 0.7522, 1. A random number of 0.6 falls in the interval of E, which becomes the second centre; the far points D, E and F together had a probability of 0.8991.

**Step 2. The silhouette of A.**  $a = (1.118034 + 3.201562)/2 = 2.159798$  from the distances to B and C, and  $b = (7.211103 + 4.716991 + 5.315073)/3 = 5.747722$  from the distances to D, E and F, so

$$s(A) = \frac{5.747722 - 2.159798}{5.747722} = 0.624234$$

**Step 3. The silhouette of C.**  $a = (3.201562 + 2.12132)/2 = 2.661441$  and  $b = (4.031129 + 1.581139 + 2.12132)/3 = 2.577863$ , so

$$s(C) = \frac{2.577863 - 2.661441}{2.661441} = -0.031403$$

C belongs to cluster 1 by its distance to the centres, yet on average it lies closer to the points of cluster 2. Its negative silhouette marks it as a point between clusters, which an expert should inspect. Part A of the [interactive lab](#) computes all these quantities for any points.

## Principal component analysis

Principal component analysis finds orthogonal directions along which the data vary most [14, 15]. The first component is the direction of largest variance, the second the largest remaining variance orthogonal to the first, and so on. The explained variance ratio of each component shows how much information a projection keeps, and the loadings show how each original variable contributes. In well logging, the first components often combine porosity and density logs, which respond to the same rock properties. PCA serves visualisation, compression, noise reduction and the detection of anomalies as points that the leading components reconstruct poorly.

### Formulas: Principal components from the covariance matrix

The  $n$  observations of  $p$  variables are first centred by subtracting each variable's mean. The covariance matrix  $S$  collects the variances on its diagonal and the covariances off it [15]:

$$S_{jl} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{il} - \bar{x}_l)$$

The principal components are the eigenvectors  $v$  of  $S$ , the directions that  $S$  only stretches,  $Sv = \lambda v$ , and the eigenvalue  $\lambda$  is the variance of the data along  $v$  [14]. For two variables the eigenvalues solve a quadratic equation in the trace  $\text{tr} = S_{11} + S_{22}$  and the determinant  $\det = S_{11}S_{22} - S_{12}^2$ :

$$\lambda_{1,2} = \frac{\text{tr} \pm \sqrt{\text{tr}^2 - 4\det}}{2}$$

and the first row of  $(S - \lambda_1 I)v = 0$  gives the direction  $v \propto (1, (\lambda_1 - S_{11})/S_{12})$ , scaled to length one. The explained variance ratio of component  $j$  is  $\lambda_j / \sum_l \lambda_l$ . A centred observation  $x$  has the score  $z = x \cdot v_1$  on the first component, its reconstruction from that component alone is  $\hat{x} = z v_1$ , and the reconstruction error  $\|x - \hat{x}\|^2$  serves as an anomaly score.

### Worked example: Two porosity logs

Five depth samples give a density porosity of 8, 9, 10, 11, 12 percent and a neutron porosity of 10.5, 11, 12, 12.5, 14 percent.

**Step 1. Centring.** The means are 10 and 12, so the centred values are -2, -1, 0, 1, 2 and -1.5, -1, 0, 0.5, 2.

**Step 2. The covariance matrix.**

$$S_{11} = \frac{4+1+0+1+4}{4} = 2.5, S_{22} = \frac{2.25+1+0+0.25+4}{4} = 1.875$$

$$S_{12} = \frac{(-2)(-1.5) + (-1)(-1) + 0 \times 0 + 1 \times 0.5 + 2 \times 2}{4} = \frac{3+1+0+0.5+4}{4} = 2.125$$

**Step 3. Eigenvalues.**  $\text{tr} = 2.5 + 1.875 = 4.375$  and  $\det = 2.5 \times 1.875 - 2.125^2 = 4.6875 - 4.515625 = 0.171875$ :

$$\lambda_{1,2} = \frac{4.375 \pm \sqrt{4.375^2 - 4 \times 0.171875}}{2} = \frac{4.375 \pm \sqrt{18.453125}}{2} = \frac{4.375 \pm 4.29571}{2}$$

so  $\lambda_1 = 4.335355$  and  $\lambda_2 = 0.039645$ .

**Step 4. The first direction.**  $(\lambda_1 - S_{11})/S_{12} = (4.335355 - 2.5)/2.125 = 0.863696$ , and the length of  $(1, 0.863696)$  is  $\sqrt{1+0.863696^2} = 1.321352$ , so  $v_1 = (0.756801, 0.653646)$ .

**Step 5. Explained variance.**  $4.335355/4.375 = 0.990938$ : The first component keeps 99.1 percent of the variance, because both logs respond to the same porosity.

**Step 6. Score and reconstruction of the last sample.** For the centred sample  $(2, 2)$ :

$$z = 2 \times 0.756801 + 2 \times 0.653646 = 2.820894$$

$$\hat{x} = 2.820894 \times (0.756801, 0.653646) = (2.134855, 1.843866)$$

$$\|x - \hat{x}\|^2 = (2 - 2.134855)^2 + (2 - 1.843866)^2 = 0.042564$$

A sample whose logs disagree, for example because of a washed-out borehole, would lie far from the first direction and show a much larger error. Part B of the [interactive lab](#) computes these steps for any data.

### Density-based clustering

Clusters in engineering data are not always round. Earthquakes line up along faults, defects form streaks on a web of fabric, and traffic accidents gather along roads. DBSCAN defines clusters as regions of high

density: A point with at least a minimum number of neighbours within a radius  $\epsilon$  is a core point, clusters grow by connecting core points that are neighbours, and points that belong to no cluster are labelled noise [5]. DBSCAN finds clusters of arbitrary shape and does not need the number of clusters, but  $\epsilon$  and the minimum number of points must fit the density of the data. For epicentres, distances should be measured on the sphere, for example with the haversine formula, rather than in degrees.

**Check your understanding.** Aftershocks form an elongated band along a fault, surrounded by scattered background events. Which method suits this pattern best?

- A. k-means with  $k = 2$
- B. DBSCAN
- C. PCA with one component
- D. Linear regression

*Answer: B. DBSCAN follows dense regions of any shape and labels sparse background events as noise [5].*

### Formulas: DBSCAN and distances on the sphere

DBSCAN uses two parameters, a radius  $\epsilon$  and a minimum number of points  $m$  [5]. The  $\epsilon$ -neighbourhood of a point  $p$ , which includes  $p$  itself, is

$$N_{\epsilon}(p) = \{q : d(p, q) \leq \epsilon\}$$

A point is a core point when  $|N_{\epsilon}(p)| \geq m$ . A cluster is grown from a core point by adding its neighbours and, recursively, the neighbours of every core point among them. A point that is not a core point but lies in the neighbourhood of one is a border point, and every remaining point is noise. Epicentres are given by latitude  $\phi$  and longitude  $\lambda$  on a sphere of radius  $R = 6371$  km, and the great-circle distance between two of them follows from the haversine formula, with all angles in radians:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right), d = 2R \arcsin \sqrt{a}$$

A degree of latitude is always  $2\pi R/360 = 111.195$  km long, while a degree of longitude shrinks with  $\cos\phi$ , so distances computed directly in degrees are wrong in the east-west direction.

### Worked example: Core, border and noise points

Eight points on a map, in kilometres, are clustered with  $\epsilon = 1.2$  km and  $m = 3$ . The distance between P1 and P4 is  $\sqrt{1^2 + 1^2} = 1.414 > 1.2$ , so they are not neighbours, while P4 and P5 are 1 km apart:

| Point | Coordinates | Neighbourhood $N_{\epsilon}$ | Size | Type   |
|-------|-------------|------------------------------|------|--------|
| P1    | (0, 0)      | P1, P2, P3                   | 3    | core   |
| P2    | (1, 0)      | P1, P2, P4                   | 3    | core   |
| P3    | (0, 1)      | P1, P3, P4                   | 3    | core   |
| P4    | (1, 1)      | P2, P3, P4, P5               | 4    | core   |
| P5    | (2, 1)      | P4, P5                       | 2    | border |
| P6    | (5, 5)      | P6, P7                       | 2    | noise  |
| P7    | (5.5, 5)    | P6, P7                       | 2    | noise  |
| P8    | (9, 0)      | P8                           | 1    | noise  |

P1 to P4 are core points and form one cluster. P5 has only two points in its neighbourhood but lies next to the core point P4, so it joins the cluster as a border point. P6 and P7 are close to each other, but neither has three points in its neighbourhood, so both are noise, as is the isolated P8. With  $m = 2$ , P6 and P7

would form a second cluster: The parameters must match the density of the data. Part C of the [interactive lab](#) places points by clicking and compares k-means with DBSCAN on the same data.

### Worked example: The distance between two epicentres

Two epicentres lie at  $37.2^\circ$  N,  $37.0^\circ$  E and  $38.0^\circ$  N,  $37.2^\circ$  E.

**Step 1. Differences in radians.**  $\Delta\varphi = 0.8 \times \pi/180 = 0.013963$  and  $\Delta\lambda = 0.2 \times \pi/180 = 0.003491$ .

**Step 2. The haversine term.**

$$\sin^2(0.013963/2) = 0.0000487406, \sin^2(0.003491/2) = 0.0000030468$$

$$\cos 37.2^\circ = 0.79653, \cos 38^\circ = 0.788011$$

$$a = 0.0000487406 + 0.79653 \times 0.788011 \times 0.0000030468 = 0.000050653$$

**Step 3. The distance.**  $2 \arcsin \sqrt{0.000050653} = 0.0142343$ , so  $d = 6371 \times 0.0142343 = 90.69$  km.

**Step 4. The error of degrees.** Treating degrees as a flat grid gives  $\sqrt{0.8^2 + 0.2^2} \times 111.195 = 91.69$  km, close here because the separation is mostly north-south. For two points one degree apart in longitude at  $38^\circ$  N, however, the true distance is about  $111.195 \times \cos 38^\circ = 87.62$  km, while degrees would give 111.195 km, an error of 27 percent that would distort every  $\varepsilon$ -neighbourhood. Part D of the [interactive lab](#) clusters epicentres with haversine distances.

### Signals in the frequency domain

Many engineering sensors produce time series sampled at a fixed rate: accelerometers on machines, microphones, current transducers, seismometers. The discrete Fourier transform expresses a sampled signal as a sum of sinusoids, and the fast Fourier transform computes it efficiently [6]. The sampling rate  $f_s$  limits the highest frequency that can be represented to  $f_s/2$ , the Nyquist frequency, and a record of duration  $T$  resolves frequencies that differ by  $1/T$ . Rotating machines produce peaks at the shaft frequency and its harmonics, and faults add their own frequencies.

A rolling-element bearing with a damaged outer race produces an impact every time a rolling element passes the damage. The ball pass frequency of the outer race is  $\text{BPFO} = (n/2) f_r (1 - (d/D) \cos \phi)$ , with  $n$  rolling elements of diameter  $d$ , pitch diameter  $D$ , contact angle  $\phi$  and shaft frequency  $f_r$ . The impacts excite structural resonances at high frequency, so the fault frequency often appears not in the raw spectrum but in the spectrum of the signal's envelope, obtained after demodulation. Envelope analysis is the benchmark technique of bearing diagnostics [7], and the Case Western Reserve University data are the standard test set for it [16]. Simple condition indicators complement spectra: the root mean square (RMS) measures vibration energy, kurtosis measures impulsiveness, and the crest factor compares the peak with the RMS.

**Check your understanding.** A bearing has 9 balls,  $d/D = 0.2034$ , contact angle zero, and the shaft turns at 30 Hz. What is the BPFO?

- A. 30 Hz
- B. About 107.5 Hz
- C. About 135 Hz
- D. About 162.5 Hz

*Answer: B.  $\text{BPFO} = 4.5 \times 30 \times (1 - 0.2034)$ , which is about 107.5 Hz.*

### Formulas: The discrete Fourier transform, bearing frequencies and condition indicators

A signal sampled at the rate  $f_s$  gives  $N$  values  $x_0, \dots, x_{N-1}$  over the duration  $T = N/f_s$ . The discrete Fourier transform writes it as a sum of complex sinusoids with the coefficients [6]

$$X_k = \sum_{n=0}^{N-1} x_n e^{-i2\pi kn/N}, k=0,1,\dots,N-1$$

where  $i$  is the imaginary unit and  $e^{-i\theta} = \cos \theta - i \sin \theta$ . Coefficient  $k$  belongs to the frequency  $f_k = k f_s / N$ , so neighbouring bins are  $\Delta f = f_s / N = 1/T$  apart, and the highest frequency that can be represented is the Nyquist frequency  $f_s / 2$ . The mean of the signal is  $X_0 / N$ , and a sinusoid that falls exactly on bin  $k$  has the amplitude  $2 |X_k| / N$ . For a bearing with  $n$  rolling elements of diameter  $d$  on the pitch diameter  $D$ , with contact angle  $\phi$  and shaft frequency  $f_r$ , the characteristic frequencies are [7]

$$\text{BPFO} = \frac{n}{2} f_r \left( 1 - \frac{d}{D} \cos \phi \right), \text{BPFI} = \frac{n}{2} f_r \left( 1 + \frac{d}{D} \cos \phi \right)$$

$$\text{FTF} = \frac{f_r}{2} \left( 1 - \frac{d}{D} \cos \phi \right), \text{BSF} = \frac{D}{2d} f_r \left( 1 - \left( \frac{d}{D} \cos \phi \right)^2 \right)$$

for the outer race, the inner race, the cage and the spin of a rolling element. The impacts at these rates modulate a high-frequency resonance; envelope analysis takes the magnitude of the analytic signal,  $|x(t) + iH\{x\}(t)|$ , where  $H$  denotes the Hilbert transform, and the fault frequencies appear in the spectrum of this envelope. Three simple indicators summarise a record:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_n x_n^2}, \text{crest factor} = \frac{\max_n |x_n|}{\text{RMS}}, \text{kurtosis} = \frac{\frac{1}{N} \sum_n (x_n - \bar{x})^4}{\left( \frac{1}{N} \sum_n (x_n - \bar{x})^2 \right)^2}$$

A Gaussian signal has a kurtosis of 3; impacts raise it.

### Worked example: A four-point Fourier transform

The samples  $x = (2, 1, 0, 1)$  are taken at  $f_s = 1000$  Hz.

**Step 1. The rotating factors.** For  $k=1$  and  $N=4$ ,  $e^{-i2\pi n/4}$  takes the values  $1, -i, -1$  and  $i$  for  $n=0, 1, 2, 3$ ; for  $k=2$  the values  $1, -1, 1, -1$ ; for  $k=3$  the values  $1, i, -1, -i$ .

**Step 2. The coefficients.**

$$X_0 = 2 + 1 + 0 + 1 = 4, X_1 = 2 \times 1 + 1 \times (-i) + 0 \times (-1) + 1 \times i = 2$$

$$X_2 = 2 - 1 + 0 - 1 = 0, X_3 = 2 + i + 0 - i = 2$$

**Step 3. Interpretation.** The bins are  $\Delta f = 1000/4 = 250$  Hz apart. The mean is  $X_0/4 = 1$ , and bin 1, at 250 Hz, holds a sinusoid of amplitude  $2 \times 2/4 = 1$ : Indeed  $x_n = 1 + \cos(2\pi n/4)$ . Bin 3 mirrors bin 1, as for every real signal, and lies above the Nyquist frequency of 500 Hz. Part E of the [interactive lab](#) computes the Fourier coefficients of any short signal term by term.

### Worked example: Bearing frequencies and indicators

**Step 1. Fault frequencies.** For  $n=9$ ,  $d/D=0.2034$ ,  $\phi=0$  and  $f_r=30$  Hz, as in the knowledge check:

$$\text{BPFO} = 4.5 \times 30 \times (1 - 0.2034) = 135 \times 0.7966 = 107.541 \text{ Hz}, \text{BPFI} = 135 \times 1.2034 = 162.459 \text{ Hz}$$

$$\text{FTF} = 15 \times 0.7966 = 11.949 \text{ Hz}, \text{BSF} = \frac{30}{2 \times 0.2034} \times (1 - 0.2034^2) = 70.695 \text{ Hz}$$

At the 1797 rpm of the Case Western Reserve data,  $f_r = 1797/60 = 29.95$  Hz, and the BPFO is  $4.5 \times 29.95 \times 0.7966 = 107.36$  Hz [16].

**Step 2. Resolution.** A record sampled at 12 kHz for one second gives  $\Delta f = 12000/12000 = 1$  Hz, fine enough to separate BPFO from its neighbours, and a Nyquist frequency of 6000 Hz.

**Step 3. Indicators of a short record.** The samples 0.5, -0.4, 0.3, 3, -0.6, 0.4, -0.5, 0.3 contain one impact. Their squares add up to 10.36, so  $\text{RMS} = \sqrt{10.36/8} = 1.137981$  and the crest factor is  $3/1.137981 = 2.636248$ . The mean is  $3/8 = 0.375$ ; the squared deviations from it add up to 9.235, so the variance is 1.154375, and the fourth powers add up to 49.331641, so  $\frac{1}{8} \sum (x - \bar{x})^4 = 6.166455$  and

$$\text{kurtosis} = \frac{6.166455}{1.154375^2} = 4.6275$$

well above 3, which flags the impulsive content. Part F of the [interactive lab](#) computes the frequencies for any bearing and the indicators for any record, and Part G synthesises the vibration of a bearing and shows its waveform, spectrum and envelope spectrum.

## Anomaly detection

An anomaly detector learns what normal data look like and scores how unusual a new record is [2]. Statistical rules flag values far from the mean in units of standard deviation. Reconstruction-based methods flag records that a PCA model or an autoencoder reconstructs poorly. Isolation forests grow random trees that split the data at random thresholds; anomalies are isolated after few splits, so the average path length to isolate a point becomes its score [8]. In condition monitoring, detectors are usually trained on data from normal operation, because faults are rare and diverse. The alarm threshold, as in Week 8, depends on the costs of false alarms and missed faults, and a detector must be re-examined when operating conditions change, since a new load or speed can look like an anomaly [17].

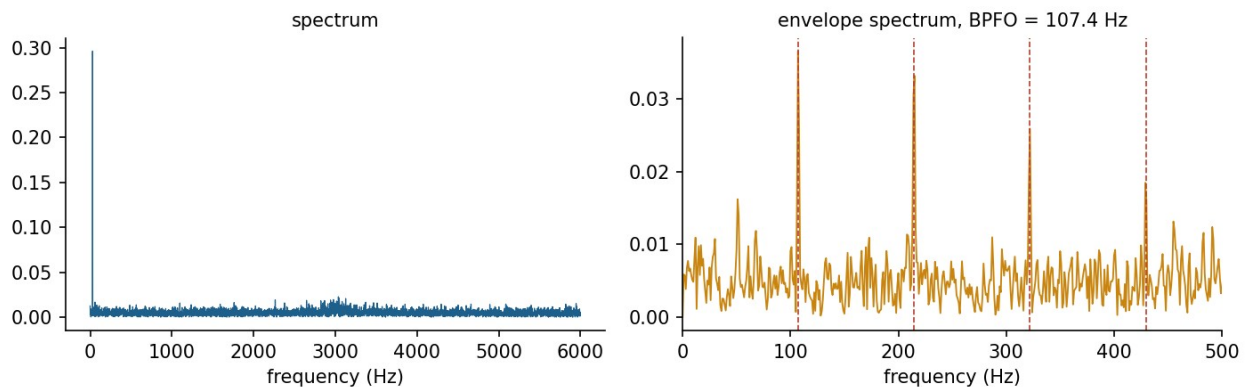


Figure 9.2. Spectrum and envelope spectrum of a synthetic vibration signal from a bearing with an outer-race defect. The fault frequency and its harmonics stand out only in the envelope spectrum [7].

**Check your understanding.** An isolation forest was trained on vibration features from a pump at 50 percent load. After the load is raised to 90 percent, alarms rise sharply although the pump is healthy. What is the most likely explanation?

- A. Isolation forests cannot handle more than one feature
- B. The new operating condition lies outside the normal data used for training
- C. The FFT is wrong at high load
- D. Kurtosis cannot be computed at high load

**Answer:** B. The detector has learned normal behaviour for one condition; data from other conditions look unusual even without a fault.

## Formulas: Statistical rules and the isolation forest

A simple rule flags a value  $x$  whose standardised distance from the mean of normal data exceeds a limit, often 3:  $|z| = |x - \mu|/\sigma > 3$ . Reconstruction-based methods flag records with a large error  $\|x - \hat{x}\|^2$ , as in the PCA example. An isolation forest grows random trees on subsamples of  $\psi$  points, splitting on a random feature at a random value, and records the number of edges  $h(x)$  from the root to the leaf that isolates a point. Its anomaly score normalises the mean path length over all trees by the average path length  $c(\psi)$  of an unsuccessful search in a binary tree [8]:

$$s(x) = 2^{-E[h(x)]/c(\psi)}, c(\psi) = 2H(\psi - 1) - \frac{2(\psi - 1)}{\psi}$$

with the harmonic number  $H(i) \approx \ln i + 0.5772156649$ , where the constant is Euler's constant. Scores close to 1 mark anomalies, scores well below 0.5 normal points, and a score of 0.5 means that  $E[h(x)] = c(\psi)$ .

### Worked example: Two detectors

**Step 1. A three-sigma rule.** A bearing temperature has  $\mu = 70^\circ\text{C}$  and  $\sigma = 2.5^\circ\text{C}$  in normal operation. A reading of  $78.5^\circ\text{C}$  gives  $z = (78.5 - 70)/2.5 = 8.5/2.5 = 3.4$ , beyond 3, so it is flagged.

**Step 2. The normalising constant.** With subsamples of  $\psi = 256$  points, as recommended by Liu and colleagues:

$$H(255) = \ln 255 + 0.5772156649 = 5.541264 + 0.577216 = 6.118479$$

$$c(256) = 2 \times 6.118479 - \frac{2 \times 255}{256} = 12.236958 - 1.992188 = 10.24477$$

**Step 3. Two scores.** A point isolated after 4 splits on average scores  $2^{-4/10.24477} = 2^{-0.390443} = 0.762895$ , an anomaly. A point that needs 14 splits scores  $2^{-14/10.24477} = 2^{-1.366551} = 0.387817$ , a normal point. Few splits isolate a point when it lies far from the others, which is the idea of the method. Part H of the [interactive lab](#) isolates points with random splits.

## Python step 9: Signals, features and clustering with NumPy

### Sampling a signal

A vibration sensor records values at a fixed sampling rate, for example 2000 samples per second. The time stamps follow from `np.arange(n) / rate`, and a sinusoid of frequency  $f$  is `np.sin(2 * np.pi * f * t)`. The practice signal below adds a component at the running speed of a shaft, a weaker component near a bearing defect frequency and random noise.

```
import numpy as np

rate_hz = 2000                                # samples per second
time_s = np.arange(2000) / rate_hz           # one second of time
stamps
shaft = 0.8 * np.sin(2 * np.pi * 25 * time_s) # 25 Hz running speed
fault = 0.3 * np.sin(2 * np.pi * 107 * time_s) # component near a
defect frequency
noise = 0.2 * np.random.default_rng(3).normal(size=time_s.size)
vib = shaft + fault + noise
print(time_s.size, time_s[:3], round(float(vib.std()), 3))
```

Output

```
2000 [0.      0.0005 0.001 ] 0.644
```

### Statistics of a signal

Condition monitoring starts with simple statistics. The root mean square (RMS) measures the energy of a signal, the peak is its largest absolute value, and the crest factor, the peak divided by the RMS, rises when short impacts appear, which is typical of developing bearing damage.

```
rms = np.sqrt(np.mean(vib ** 2))
peak = np.max(np.abs(vib))
print(f"RMS {rms:.3f}, peak {peak:.3f}, crest factor {peak / rms:.2f}")
```

*Output*

```
RMS 0.644, peak 1.525, crest factor 2.37
```

### Windows and the axis argument

Monitoring systems compute features for short windows of a signal. `reshape` rearranges the 2000 samples into a table of 10 windows with 200 samples each; the value -1 lets NumPy work out that length. Functions such as `np.mean` accept an `axis` argument: `axis=1` works along each row and returns one value per window, and `axis=0` works down the columns.

```
windows = vib.reshape(10, -1) # 10 windows of 0.1 s each
rms_per_window = np.sqrt(np.mean(windows ** 2, axis=1))
print(windows.shape, rms_per_window.shape)
print(rms_per_window.round(3))
```

*Output*

```
(10, 200) (10,)
[0.653 0.655 0.646 0.642 0.655 0.636 0.639 0.627 0.647 0.639]
```

### Spectra

The fast Fourier transform decomposes a signal into sinusoids and shows at which frequencies its energy lies. For real signals, `np.fft.rfft` returns the non-negative frequencies, and `np.fft.rfftfreq` gives the frequency of each value. Dividing the magnitudes by half the number of samples turns them into amplitudes. A boolean mask restricts the search to a frequency band, and `np.isclose` compares floats with a tolerance instead of exact equality.

```
spectrum = np.abs(np.fft.rfft(vib)) / (vib.size / 2)
freqs = np.fft.rfftfreq(vib.size, d=1 / rate_hz)
print(freqs[np.argmax(spectrum)], "Hz is the strongest component")
band = (freqs > 80) & (freqs < 150)
print(freqs[band][np.argmax(spectrum[band])], "Hz is the strongest component between 80 and 150 Hz")
print(f"amplitude at 107 Hz: {spectrum[np.isclose(freqs, 107)][0]:.3f}")
```

*Output*

```
25.0 Hz is the strongest component
107.0 Hz is the strongest component between 80 and 150 Hz
amplitude at 107 Hz: 0.300
```

The envelope analysis of this week adds one step for bearing faults, whose short impacts spread their energy over many frequencies and appear clearly only after demodulation.

## Features for many windows at once

Unsupervised methods work on feature tables with one row per example. The code below simulates 20 windows of a healthy machine and 20 windows of a damaged one whose bearing produces short impacts. Broadcasting does most of the work: The sine wave of shape (200,) is added to noise of shape (20, 200), and NumPy repeats the sine for every row. Features computed along `axis=1` give one value per window, and `np.column_stack` joins them into a table.

```
gen9 = np.random.default_rng(9)
t_w = np.arange(200) / rate_hz                                # windows
of 0.1 s
healthy = 0.5 * np.sin(2 * np.pi * 25 * t_w) + 0.1 *
gen9.normal(size=(20, 200))
damaged = 0.5 * np.sin(2 * np.pi * 25 * t_w) + 0.1 *
gen9.normal(size=(20, 200))
damaged[:, ::40] += 1.5                                       # an impact
every 20 ms
win = np.vstack([healthy, damaged])                            # 40
windows
rms_w = np.sqrt(np.mean(win ** 2, axis=1))
crest_w = np.max(np.abs(win), axis=1) / rms_w
F = np.column_stack([rms_w, crest_w])                          # feature
table, one row per window
print(win.shape, F.shape)
print(F[[0, 1, 38, 39]].round(2))
```

Output

```
(40, 200) (40, 2)
[[0.37  1.93]
 [0.37  1.97]
 [0.44  3.71]
 [0.44  3.76]]
```

## k-means with broadcasting

k-means alternates two steps: Every point joins its nearest centre, and every centre moves to the mean of its points. The features are first standardised column by column, so that RMS and crest factor count equally. The distances from all 40 points to both centres come from a single broadcast: `Z[:, None, :]` has shape (40, 1, 2), `centres[None, :, :]` has shape (1, 2, 2), and their difference has shape (40, 2, 2). Summing over the last axis gives a (40, 2) table of squared distances, and `argmin(axis=1)` picks the nearest centre for every point.

```
Z = (F - F.mean(axis=0)) / F.std(axis=0)                      #
standardise each column
centres = Z[[0, -1]].copy()                                   # start
from the first and the last window
for _ in range(10):
    d = ((Z[:, None, :] - centres[None, :, :]) ** 2).sum(axis=2)
    cluster = d.argmin(axis=1)
    centres = np.array([Z[cluster == j].mean(axis=0) for j in
range(2)])
print(d.shape, np.bincount(cluster))
print("clusters of the healthy windows:", cluster[:20])
print("clusters of the damaged windows:", cluster[20:])
```

Output

```
(40, 2) [20 20]
clusters of the healthy windows: [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0]
clusters of the damaged windows: [1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1]
```

### A simple anomaly flag

When examples of normal behaviour are available, a z-score flags values that lie far from them:  $z = (x - \text{mean}) / \text{standard deviation}$ , with the mean and the standard deviation taken from the healthy windows only. Windows with an absolute z above 3 are reported. The anomaly detection methods of this week extend the same idea to several features at once.

```
mu, sd = rms_w[:20].mean(), rms_w[:20].std()
z = (rms_w - mu) / sd
print("flagged windows:", np.flatnonzero(np.abs(z) > 3))
```

*Output*

```
flagged windows: [20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37
38 39]
```

**Check your understanding.** For `a = np.arange(12).reshape(3, 4)`, what is the shape of `a.sum(axis=0)`?

- A. (3,)
- B. (4,)
- C. (12,)
- D. (3, 4)

*Answer: B. axis=0 sums down the rows, which leaves one value per column, and the array has four columns.*

### Discipline challenges

Unlabelled data are everywhere. Choose a row, apply the method named there, and ask an expert, or the literature, whether the structure found makes sense.

| Department                                            | Challenge                                                                                                                  |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Geological Engineering and Earth Sciences Engineering | Cluster the SEG well logs with and without the PE log and compare the clusters with the facies [3].                        |
| Geophysical Engineering                               | DBSCAN on the USGS catalogue for a region and period of your choice; compare clusters with known fault zones [4, 18].      |
| Mechanical Engineering and Automotive Engineering     | Envelope analysis and kurtosis for bearing signals; test the detector at a new shaft speed [7].                            |
| Electrical and Electronics Engineering                | Load profiles of buildings or feeders clustered into daily patterns with k-means.                                          |
| Civil Engineering                                     | Anomaly detection on strain or acceleration records of a structure after removing temperature effects.                     |
| Environmental Engineering                             | Clusters of air-quality stations by their pollutant time profiles; anomalies as sensor faults or pollution events.         |
| Mining Engineering                                    | Clusters of seismic events in a mine and their relation to excavation fronts, with the seismic-bumps data as a start [19]. |

| Department                         | Challenge                                                                                                         |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Food Engineering and Biology       | PCA of spectroscopic or compositional measurements to separate product origins.                                   |
| Chemistry and Chemical Engineering | PCA-based monitoring of a batch process: reconstruction error as an alarm.                                        |
| Physics and Statistics             | Compare k-means, a Gaussian mixture and DBSCAN on simulated data with known structure; discuss what each assumes. |

## Weekly task

Apply clustering or anomaly detection to data from your department, either from the notebook or from a public source in DATASETS.md. Justify the scaling, the method and its parameters, show one figure that an expert of your field could check, and state what the result would need before it could support a decision. Write about 500 words and cite at least three works from this week's references [2, 5, 13].

## Research and report assignment (optional)

**Unsupervised learning in monitoring.** Review unsupervised or semi-supervised methods for monitoring in one domain, such as machine condition monitoring, seismology, structural health monitoring or process control. Compare how normal behaviour is defined, how thresholds are set, how results are validated without labels and how changes of operating conditions are handled [2, 17, 18].

Both assignments are optional. During an active semester, the weekly task can be sent together with the learning log exported from the interactive lab, and the research report on its own, to [utkukose@sdu.edu.tr](mailto:utkukose@sdu.edu.tr) or [utkukose@gmail.com](mailto:utkukose@gmail.com). They carry no separate weight, but the instructor may take them into account as a discretionary adjustment of the midterm component of the grade. The assessment section of the syllabus gives the details, and research reports follow the report template of the course repository.

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